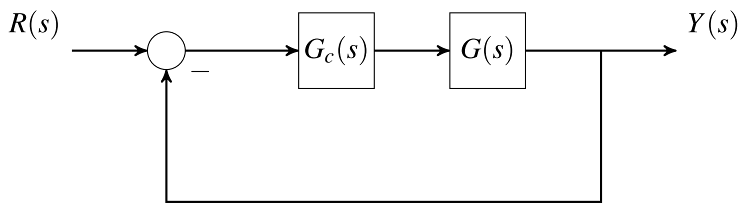
⇒ Need to be fully done with Final Exam Rev early

ECE 4375, Exam # 2 Review Material

1. Given the block diagram



Perform the following tasks

- Determine the BIBO stability with the Routh Array
- Sketch the Root Locus Where appropriate find:
  - σ_i
  - Θ_k
  - breakaway, BA, and break-in, BI points
  - gain K at BA and BI points
  - critical gain for stability, K_{cr}
- Determine the steady state error for position, velocity, and acceleration

For

A.

$$G(s) = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)}$$

B.

$$G(s) = \frac{s-9}{(s-3)(s^2-6s+5)(s^2+2s+5)}$$

C.

$$G(s) = \frac{1}{(s+1)(s+10)}$$

D.

$$G(s) = \frac{s+0.1}{(s+1)(s+10)}$$

multiple same roots.

E.

$$G(s) = \frac{s+1}{s^2(s+10)^2}$$

count them correctly

F.

$$G(s) = \frac{900}{s^2+15s+900}$$

verify answer

2. Given the DC servo motor parameters:

33 pts

- $R_a = 1.1$
- $L_a = 0.0001$
- $K_m = 12$
- $B = 0.75$
- $J_t = 0.85$
- $K_b = 0.125$

1.) Kp, final answer, -10?

2.) Good

3.) -20

Grade: 60-70?

~~θ_i~~ -4

~~θK~~ -4

~~graph~~ -4

K_{BA}/K_{BI} -4

BA/BI -4

Perform

A. When the open-loop transfer function is $\frac{\omega_m}{V_a}$

1. The open-loop transfer function, $\frac{I_a}{V_a}$. What is the steady state current given a setpoint input?
2. The poles and the time constants

B. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{ref}}$

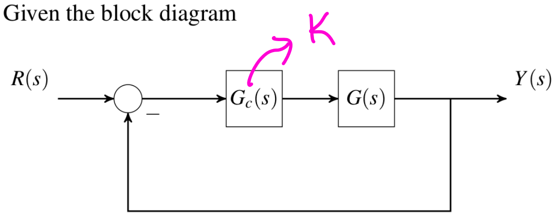
1. The open-loop transfer function, $\frac{I_a}{\Theta_{ref}}$. What is the steady state current given a setpoint input?
2. The poles and the time constants

C. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{ref}}$ *and* there is proportional gain and tachometer gain

1. Determine the range of K_p and K_f so the system is stable
2. Add a disturbance input to the model and find the open-loop transfer function $\frac{\Theta_m}{T_d}$. Determine values for K_p and K_f to minimize the disturbance

1. Given the block diagram

$G_c(s) = K = 1$



Perform the following tasks

- Determine the BIBO stability with the Routh Array
- Sketch the Root Locus Where appropriate find:
  a) – σ_i
  b) – Θ_k
  c) – breakaway, BA, and break-in, BI points
  d) – gain K at BA and BI points
  e) – critical gain for stability, K_{cr}
- Determine the steady state error for position, velocity, and acceleration

For

A.

$$G(s) = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)}$$

BIBO)

need to do it with K from now on

$$\frac{Y(s)}{R(s)} = \frac{KG}{1+GH} = \frac{\frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)}}{1+\frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)}} = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)+s^2+2s+2}$$

$$\frac{Y(s)}{R(s)} = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)+s^2+2s+2} = \frac{B(s)}{A(s)}$$

$$A(s) = (s^2-11s+24)(s^2+7s)+s^2+2s+2 = s^4-4s^3-52s^2+170s+2$$

RA

s^4	1	−52	2
s^3	−4	170	0
s^2	K_{31}	0	0
s^1	K_{41}	0	0
s^0	K_{51}	0	0

sign change

Atleast one sign change so System is unstable

Do with K

$$L = G_c(s)\,G(s)\,H(s) = K\,G(s)\,H(s) = K\,G(s)\,H(s)$$

$$G(s) = \frac{s^2+2s+2}{(s^2-11s+24)(s^2+7s)}, \quad H(s) = 1$$

$$L = \frac{K(s^2+2s+2)}{(s^2-11s+24)(s^2+7s)} = \frac{N_L}{D_L}$$

Lg zeros: $-1+j,\ -1-j$ $(m=2)$

Lg poles: $3, 8, -7, 0$ $(n=4)$

a)

$$\sigma_i = \frac{\Sigma \text{ Real part of poles} - \Sigma \text{ Real part of zeros}}{n-m} = \frac{(3+8-7)-(-1+-1)}{4-2}$$

$$\sigma_i = \frac{4-(-2)}{2} = \frac{6}{2} = 3$$

b)

$$K = \{0, \dots |n-m|-1\} = \{0, 1\}$$

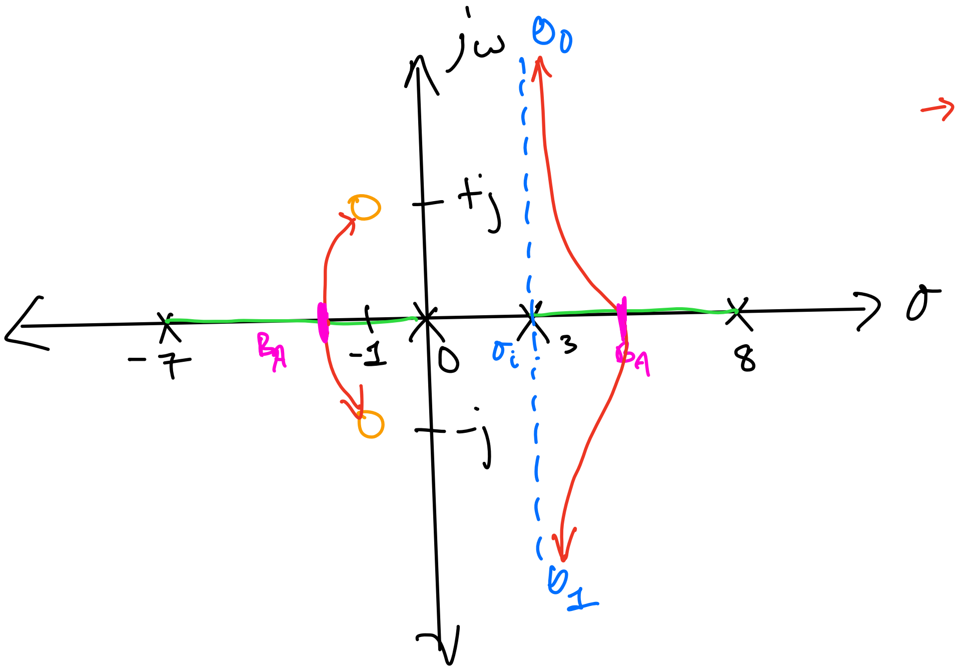
$$\theta_K = \frac{(2k+1)\pi}{n-m} \qquad \theta_0 = \frac{\pi}{2}, \qquad \theta_1 = \frac{3\pi}{2}$$

c) $$L = \frac{K(s^2+2s+2)}{(s^2-11s+24)(s^2+7s)} = \frac{N_L}{D_L} \qquad L = \frac{K(s^2+2s+2)}{s^4-4s^3-53s^2+168s}$$

$$D_L N_L' - D_L' N_L = 0$$

$$(s^4-4s^3-53s^2+168s)(2s+2) - (4s^3-12s^2-106s+168)(s^2+2s+2) = 0$$

$$s = \underbrace{-1.51}_{BA},\ 0.76,\ \underbrace{5.38}_{BA},\ -2.81 \pm j4.39$$



RL path

→ For K_{CR} to exist, ↑ would need to cross the im. axis

→ After drawing sketch, visually see if a K_{CR} is possible.

D.) $K = 1$

$$K_{BA_1} = \frac{-D_L}{N_L} = \frac{-(s^4 - 4s^3 - 53s^2 + 168s)}{(s^2 + 2s + 2)}\Bigg|_{BA = -1.51} = 282.16$$

$$K_{BA_2} = \frac{-D_L}{N_L} = \frac{-(s^4 - 4s^3 - 53s^2 + 168s)}{(s^2 + 2s + 2)}\Bigg|_{BA = 5.38} = 9.96$$

E.) K_{CR}

$$L = \frac{K(s^2 + 2s + 2)}{s^4 - 4s^3 - 53s^2 + 168s} = \frac{N_L}{D_L}$$

$$A(s) = D_L + KN_L$$

$$A(s) = s^4 - 4s^3 - 53s^2 + 168s + K(s^2 + 2s + 2)$$

$$= s^4 - 4s^3 - 53s^2 + 168s + Ks^2 + 2Ks + 2K$$

$$= s^4 - 4s^3 + (-K - 53)s^2 + (168 + 2K) + 2K$$

s^4	1	$-K+53$	$2K$
s^3	-4	$168+2K$	0
s^2	X_{31}	X_{32}	0
s^1	X_{41}	X_{42}	0
s^0	X_{51}		

No K_{CR}

SS Error

$s(s-3)(s-8)(s+7)$

$$L = \frac{K(s^2+2s+2)}{s^0(s^2-11s+24)(s^2+7s)} = \frac{K(s^2+2s+2)}{s(s-3)(s-8)(s+7)}$$

Type 1

1.) $K_p = L(0) = \frac{2K}{0} \rightarrow \infty$

$$e_{K_p}(\infty) = \frac{1}{1+K_p} = \frac{1}{1+\infty} = 0$$

$$K_v = \lim_{s\to 0} s\,L(s) = \left.\frac{K(s^2+2s+2)\cancel{s}}{\cancel{s}(s-3)(s-8)(s+7)}\right|_{s=0} = \frac{2K}{168} = \frac{K}{84}$$

$$e_{K_v}(\infty) = \frac{1}{K_v} = \frac{84}{K}$$

$$K_A = \lim_{s \to 0} s^2 L(s) = \left.\frac{K(s^2+2s+2)\cancel{s^2}}{\cancel{s}(s-3)(s-8)(s+7)}\right|_{s=0} = \frac{0}{168} = 0$$

$$e_{KA} = \frac{1}{K_A} = \frac{1}{0} \rightarrow \infty$$

$\Rightarrow$ There is no steady state for an unstable system, and thus you cannot calculate a steady state error

B.

$$G(s) = \frac{s-9}{(s-3)(s^2-6s+5)(s^2+2s+5)}$$

BIBO

$$\frac{Y(s)}{R(s)} = \frac{KG}{1+GH} = \frac{K(s-9)}{(s-3)(s^2-6s+5)(s^2+2s+5)} = \frac{\frac{K(s-9)}{(s-3)(s^2-6s+5)(s^2+2s+5)}}{1+\frac{K(s-9)}{(s-3)(s^2-6s+5)(s^2+2s+5)}}$$

$$= \frac{K(s-9)}{(s-3)(s^2-6s+5)(s^2+2s+5)+K(s-9)}$$

$$= \frac{K(s-9)}{s^5-7s^4+10s^3-14s^2+85s-75+Ks-9K} = \frac{B(s)}{A(s)}$$

$$= s^5-7s^4+10s^3-14s^2+(85+K)s-9K-75 = A(s)$$

$$\left.\begin{array}{c|ccc} s^5 & 1 & 10 & (85+K) \\ s^4 & -7 & -14 & -9K-75 \end{array}\right\}$$ 1 sign change

So unstable

$$L = KGH = \frac{K(s-9)}{(s-3)(s^2-6s+5)(s^2+2s+5)}$$

lg zeros : 9 $\qquad m = 1$

lg poles: 3, 1, 5, $-1+j2$, $-1-j2$ $\qquad n = 5$

a)

$$\sigma_i = \frac{(3+1+5-1-1)-(9)}{5-1} = \frac{-2}{4} = -\frac{1}{2}$$

b) $K = \{0, \dots, |n-m|-1\} = \{0, 1, 2, 3\}$

$$\theta_K = \frac{(2k+1)\pi}{n-m} \qquad \theta_0 = \frac{\pi}{4},\ \theta_1 = \frac{3\pi}{4},\ \theta_2 = \frac{5\pi}{4},\ \theta_3 = \frac{7\pi}{4}$$

c) BA & BI

$$L = \frac{K(s-9)}{s^5 - 7s^4 + 10s^3 - 14s^2 + 85s} = \frac{N_L}{D_L}$$

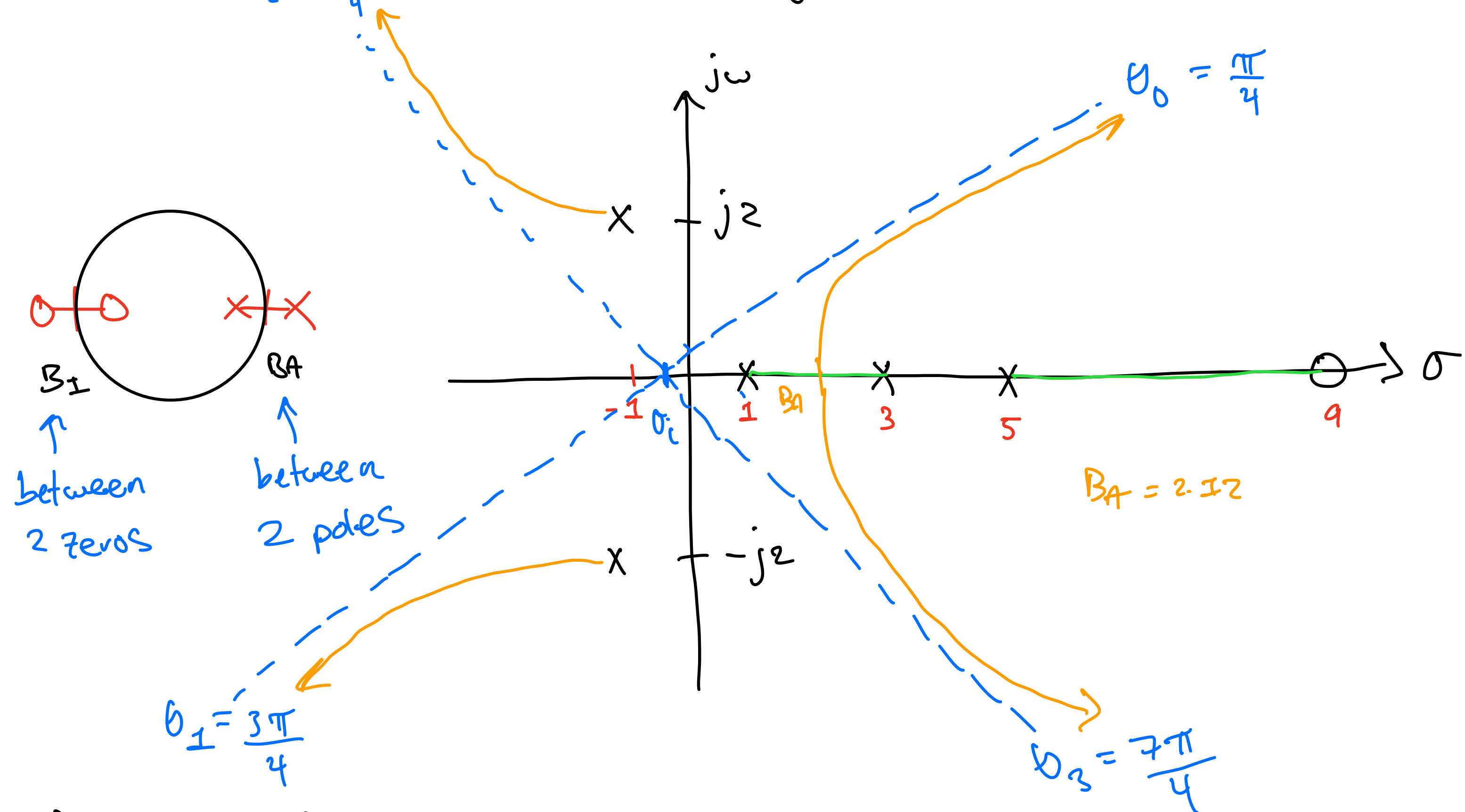
$$D_L N_L' - D_L' N_L = 0$$

$$(s^5 - 7s^4 + 10s^3 - 14s^2 + 85s)(1) - (s-9)(5s^4 - 28s^3 + 30s^2 - 28s + 85)$$

$s = -0.37 \pm j\,1.26,\ 2.12,\ 4.36,\ 10.75$

$s = \cancel{-0.37 \pm j1.26}, 2.12, \cancel{4.36}, \cancel{10.75}$ lg zeros : 9

lg poles: 3, 1, 5, $-1+j2$, $-1-j2$



D.) $K_{BA} = \dfrac{-D_L}{N_L}$

$$L = \frac{K(s-9)}{s^5 - 7s^4 + 10s^3 - 14s^2 + 85s} = \frac{N_L}{D_L}$$

$$K_{BA} = \left.\frac{-(5s^4 - 28s^3 + 30s^2 - 28s + 85)}{(s-9)}\right|_{B_A = 2.12} = 5.67$$

E.) Unstable system with no K crossing im. axis, so there's no K_{CR}

SS Error

Since unstable system,

$K_p \to 0, \; e_{Kp}(\infty) = \infty$

$K_v \to 0, \; e_{kp}(\infty) = \infty$

$K_A \to 0, \; e_{KA}(\infty) = \infty$

2. Given the DC servo motor parameters:

- $R_a = 1.1$
- $L_a = 0.0001$
- $K_m = 12$
- $B = 0.75$
- $J_t = 0.85$
- $K_b = 0.125$

step input: $v_a(t) = 1$, $V_a(s) = \frac{1}{s}$

set-point input: $V_a(t) = 1$, $V_a(s) = \frac{1}{s}$

impulse input: $V_a(t) = \delta(t)$, $V_a(s) = 1$

$v_a(t) \Leftrightarrow \theta_{ref}(t)$

$V_a(s) \Leftrightarrow \theta_{ref}(s)$

Perform

A. When the open-loop transfer function is $\frac{\omega_m}{V_a}$ ← DC Servo motor BD

1. The open-loop transfer function, $\frac{I_a}{V_a}$. What is the steady state current given a setpoint input?
2. The poles and the time constants

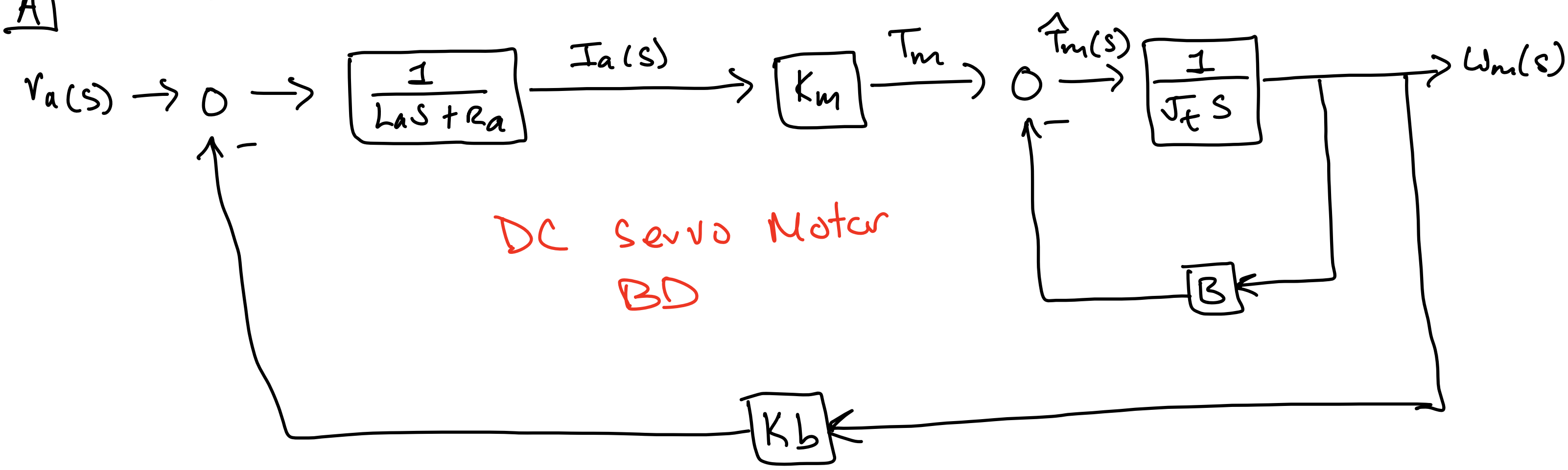
B. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{ref}}$ ← Pos. Track. sys w/ DC servo motor BD

1. The open-loop transfer function, $\frac{I_a}{\Theta_{ref}}$. What is the steady state current given a setpoint input?
2. The poles and the time constants

C. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{ref}}$ *and* there is proportional gain and tachometer gain

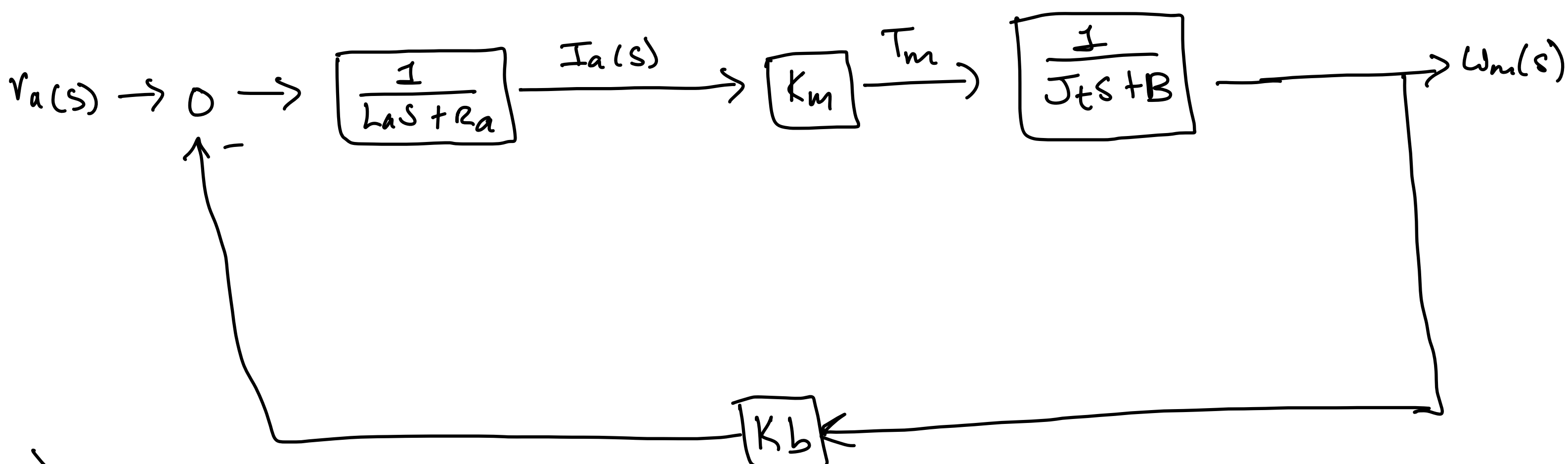
1. Determine the range of K_p and K_f so the system is stable
2. Add a disturbance input to the model and find the open-loop transfer function $\frac{\Theta_m}{T_d}$. Determine values for K_p and K_f to minimize the disturbance

A)



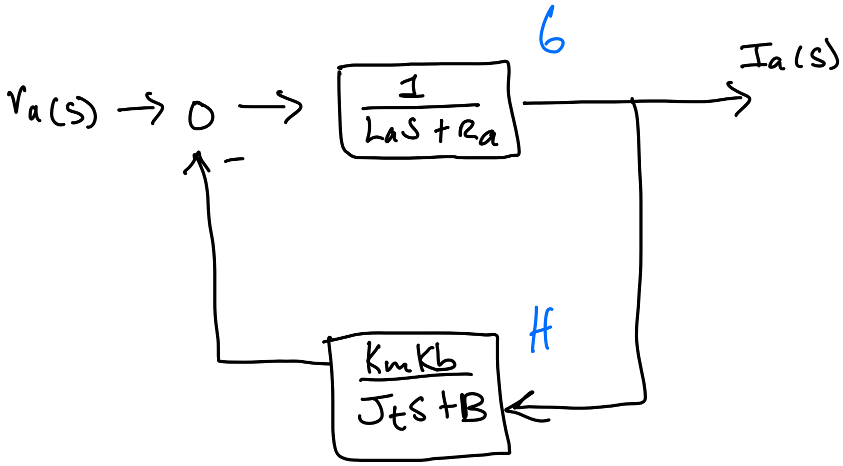
$$\frac{W_m}{T_m} = \frac{\frac{1}{J_t s}}{1 + \frac{B}{J_t s}} = \frac{1}{J_t s + B}$$

$\frac{W_m(s)}{V_a(s)} = G_m =$ DC Servo motor system



1.)

To find $\frac{I_a(s)}{V_a(s)}$, set $W_m(s) = 0$



$$\frac{I_a(s)}{V_a(s)} = \frac{\frac{1}{L_aS+R_a}}{1+\left(\frac{1}{L_aS+R_a}\right)\left(\frac{K_mK_b}{J_tS+B}\right)}$$

$$= \frac{\frac{1}{L_aS+R_a}}{1+\frac{K_mK_b}{(L_aS+R_a)(J_tS+B)}} = \frac{J_tS+B}{(L_aS+R_a)(J_tS+B)+K_mK_b}$$

$$= \frac{J_tS+B}{L_aJ_ts^2+L_aSB+R_aJ_tS+R_aB+K_mK_b}$$

$$= \frac{J_tS+B}{L_aJ_ts^2+(L_aB+R_aJ_t)S+R_aB+K_mK_b}$$

$$\frac{I_a(S)}{V_a(S)} = \frac{0.85s+0.75}{10^{-4}\cdot 0.85s^2+(10^{-4}\cdot 0.75+1.1\cdot 0.75)s+1.1\cdot 0.75+12\cdot 0.25}$$

$$\frac{I_a(s)}{V_a(s)} = \frac{0.85s+0.75}{85\times 10^{-6}s^2+0.935s+2.325}$$ OL Form

$V_a(s) \longrightarrow$ input

setpoint input $\longrightarrow V_a(s) = \frac{1}{s}$

$$I(\infty) = \lim_{s \to 0} s I_a(s)$$

$$= \lim_{s \to 0} \cancel{s} \left[\frac{0.85s + 0.75}{85 \times 10^{-6} s^2 + 0.935 s + 2.325} \right] \cancel{\frac{1}{s}} \Bigg|_{s=0}$$

$$= \frac{0.75}{2.325} = \underline{0.322}$$

2) $$\frac{I_a(s)}{V_a(s)} = \frac{0.85s + 0.75}{0.000085 s^2 + 0.935 s + 2.325}$$

$$A(s) = 0.000085 s^2 + 0.935 s + 2.325 = 0$$

$$s = -10997.5,\ -2.49 \longrightarrow \text{Poles}$$

$$\tau_n = \frac{1}{|s|}$$

$$\underline{\tau_1} = \frac{1}{|-10997.5|} = \underline{0.000091 \text{ secs}}$$

$$\underline{\tau_2} = \frac{1}{|-2.49|} = \underline{0.402 \text{ secs}}$$

Time Constants

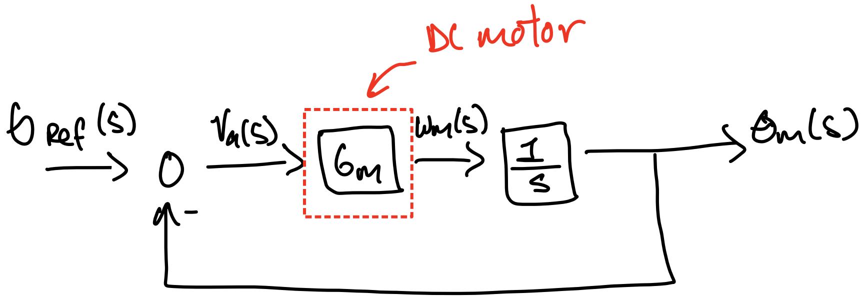
B. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{\text{ref}}}$

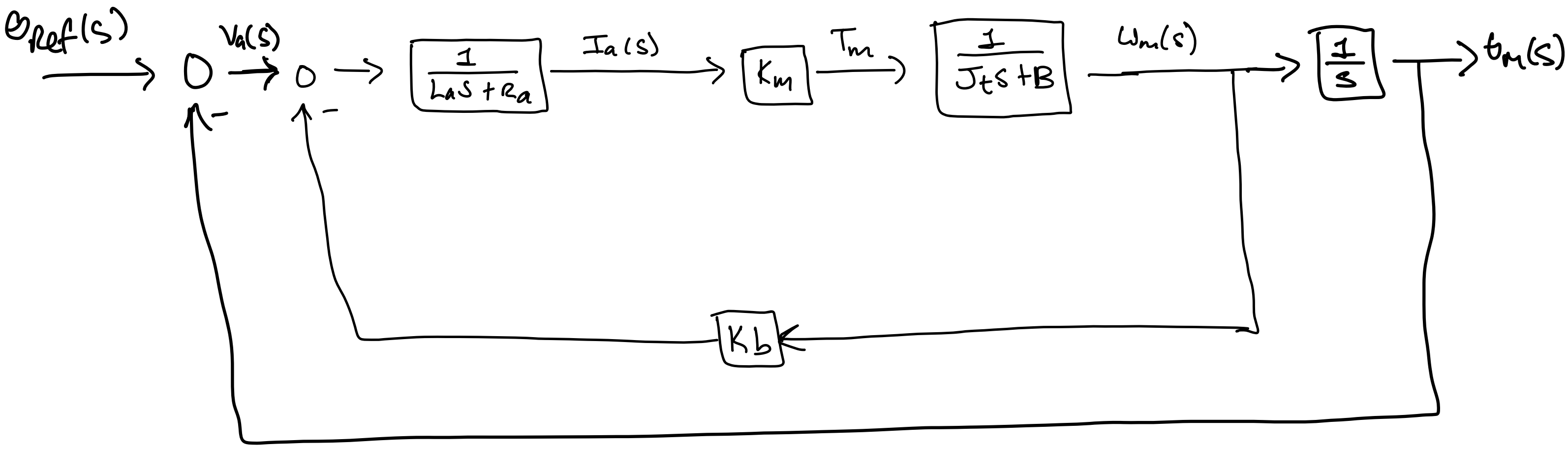
1. The open-loop transfer function, $\frac{I_a}{\Theta_{\text{ref}}}$. What is the steady state current given a setpoint input?
2. The poles and the time constants

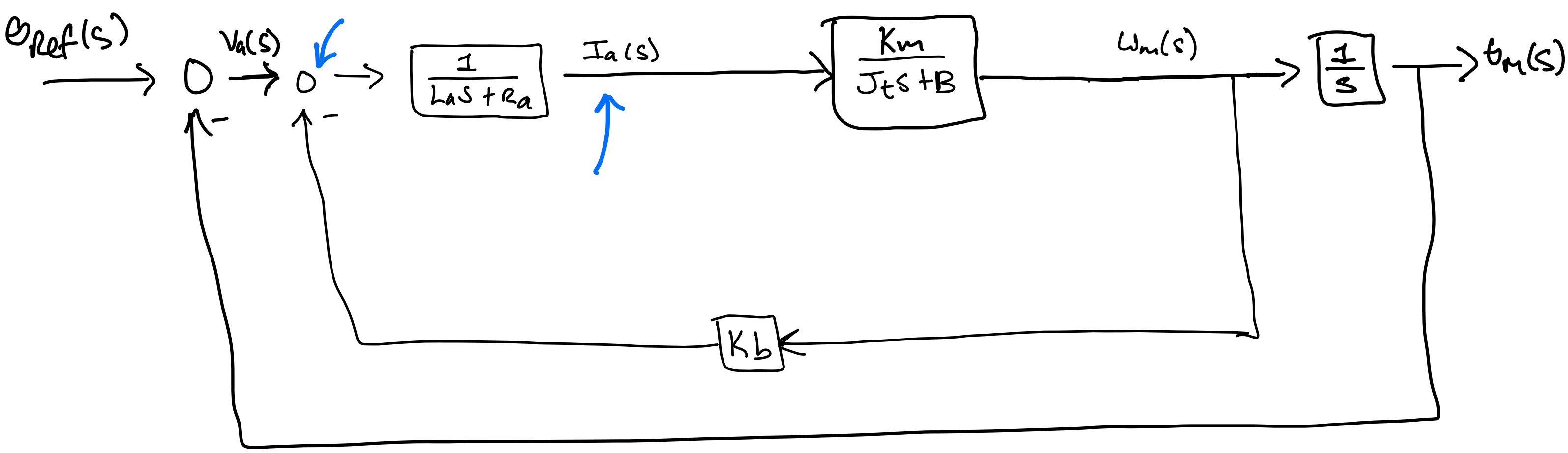
$\theta_{ref}(s) \rightarrow$ shaft Ref. Pos.

$\theta_m(s) \rightarrow$ Actual shaft pos.

Pos. Track. Sys. w/ DC servo motor

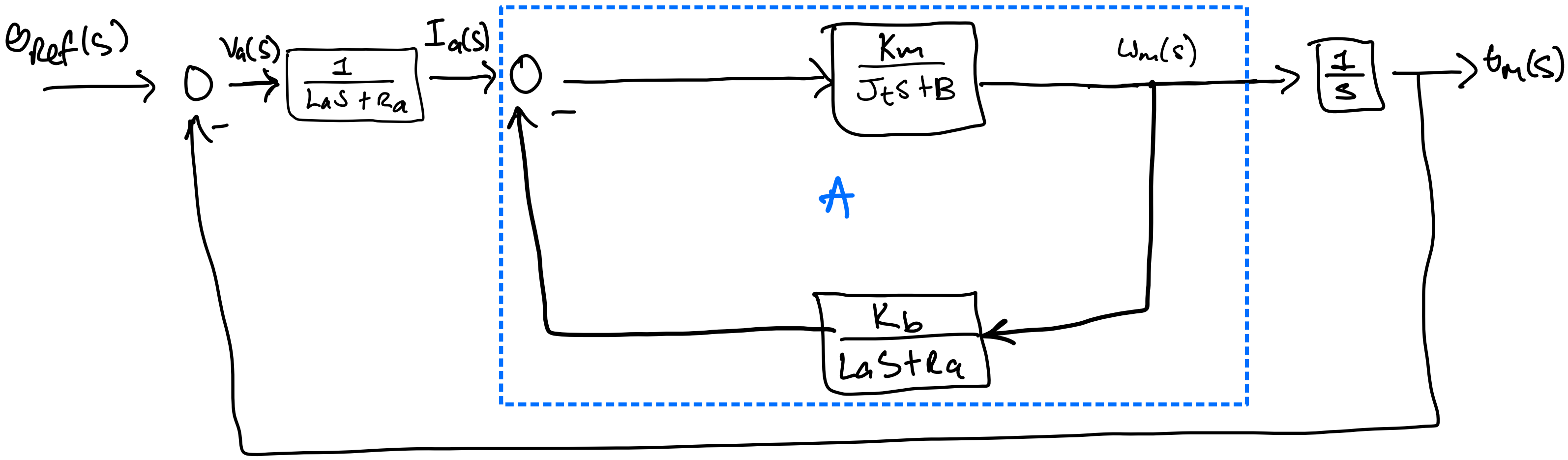






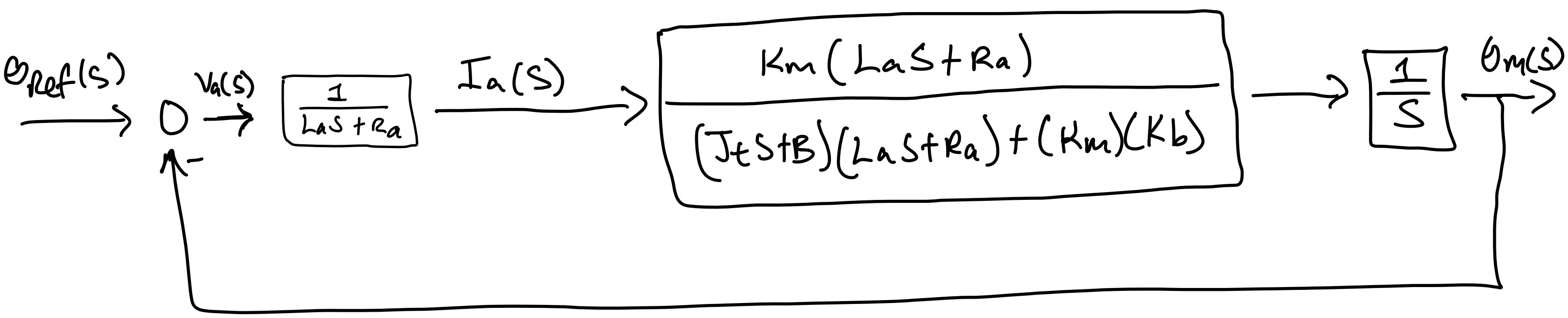
Block manipulation

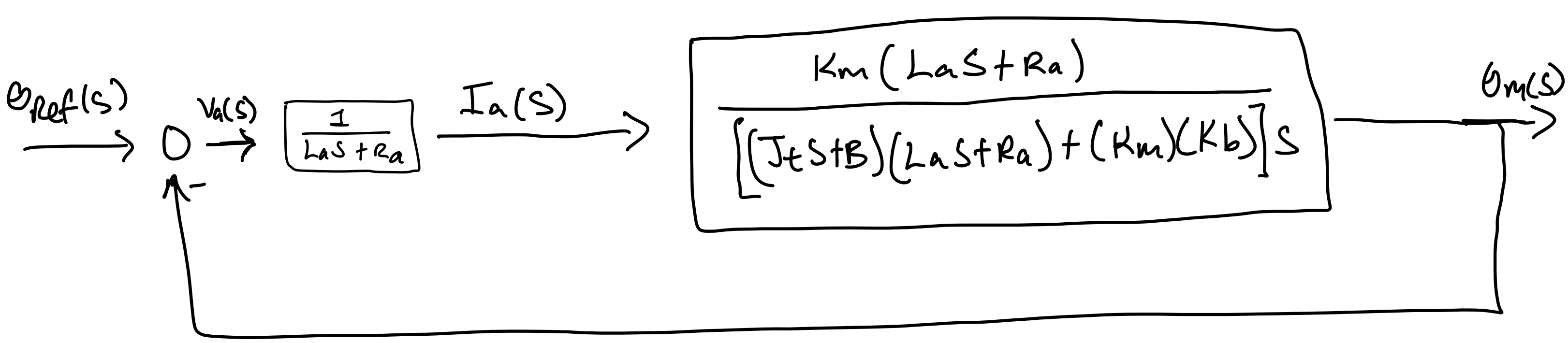
$\Rightarrow$ move summation junction after $\boxed{\frac{1}{L_a s + R_a}}$



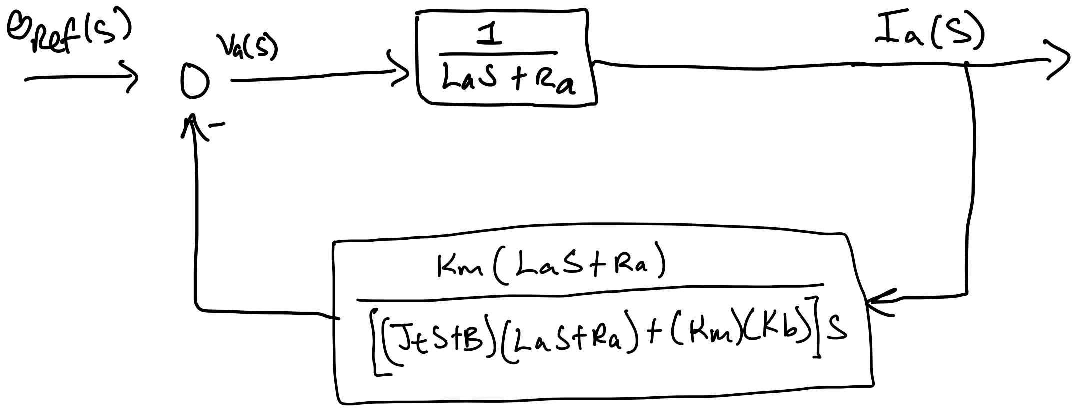
$$A = \frac{G}{1+GH} = \frac{\dfrac{K_m}{J_t s + B}}{1 + \left(\dfrac{K_m}{J_t s + B}\right)\left(\dfrac{K_b}{L_a s + R_a}\right)} = \frac{K_m}{\cancel{J_t s + B}} \cdot \frac{(\cancel{J_t s + B})(L_a s + R_a)}{(J_t s + B)(L_a s + R_a) + (K_m)(K_b)}$$

$$A = \frac{K_m(L_a s + R_a)}{(J_t s + B)(L_a s + R_a) + (K_m)(K_b)}$$





$\Rightarrow$ set $\theta_m(s) = 0$ to solve for $\dfrac{I_a(s)}{\theta_{ref}(s)}$



$$\frac{I_a(s)}{\theta_{ref}(s)} = \frac{\dfrac{1}{L_a s + R_a}}{1 + \left(\dfrac{K_m (L_a s + R_a)}{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right] s}\right)\left(\dfrac{1}{L_a s + R_a}\right)}$$

$$= \frac{\dfrac{1}{L_a \cancel{s} + R_a}}{\dfrac{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right](s)(L_a s + R_a) + K_m (L_a s + R_a)}{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right] s \,(L_a \cancel{s} + R_a)}}$$

$$= \frac{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right] s}{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right](s)(L_a s + R_a) + K_m (L_a s + R_a)}$$

$$= \frac{(J_t s^2 + Bs)(L_a s^2 + R_a) + K_m K_b s}{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right](s)(L_a s + R_a) + K_m (L_a s + R_a)}$$

$$= \frac{(J_t s^2 + Bs)(L_a s^2 + R_a) + K_m K_b s}{\left[(J_t s + B)(L_a s + R_a) + (K_m)(K_b)\right](L_a s^2 + R_a s) + K_m L_a s + K_m R_a}$$

$$= \frac{(0.85 s^2 + 0.75 s)(0.0001 s^2 + 1.1) + (12)(0.125) s}{\left[(0.85 s + 0.75)(0.0001 s + 1.1) + (12)(0.125)\right](0.0001 s^2 + 1.1 s) + 12(0.0001) s + 1.1(12)}$$

$$\frac{I_a(s)}{\theta_{Ref}(s)} = \frac{0.000085 s^4 + 0.935 s^2 + 0.000075 s^3 + 2.325 s}{0.0000000085 s^4 + 0.000187007 5 s^3 + 1.028815 s^2 + 2.5587 s + 13.2}$$

$\theta_{Ref}(s) \longrightarrow$ input

$\theta_{Ref}(s) \longrightarrow$ set point input $= \frac{1}{s}$

$$I_a(\infty) = \lim_{s \to 0} s I_a(s) =$$

$$= \lim_{s \to 0} \cancel{s} \left[\frac{0.000085 s^4 + 0.935 s^2 + 0.000075 s^3 + 2.325 s}{0.0000000085 s^4 + 0.000187007 5 s^3 + 1.028815 s^2 + 2.5587 s + 13.2}\right] \frac{1}{\cancel{s}} =$$

$$\left.\frac{0.000085 s^4 + 0.935 s^2 + 0.000075 s^3 + 2.325 s}{0.0000000085 s^4 + 0.000187007 5 s^3 + 1.028815 s^2 + 2.5587 s + 13.2}\right|_{s=0} =$$

$$= \frac{0}{13.2} = 0$$

$$\underline{I_a(\infty) = 0}$$

$$A(s) = 0.0000000085 s^4 + 0.0001870075 s^3 + 1.028815 s^2 + 2.5587 s + 13.2$$

$$s = -11016.6,\ -10981.77 \quad \rightsquigarrow \text{poles}$$

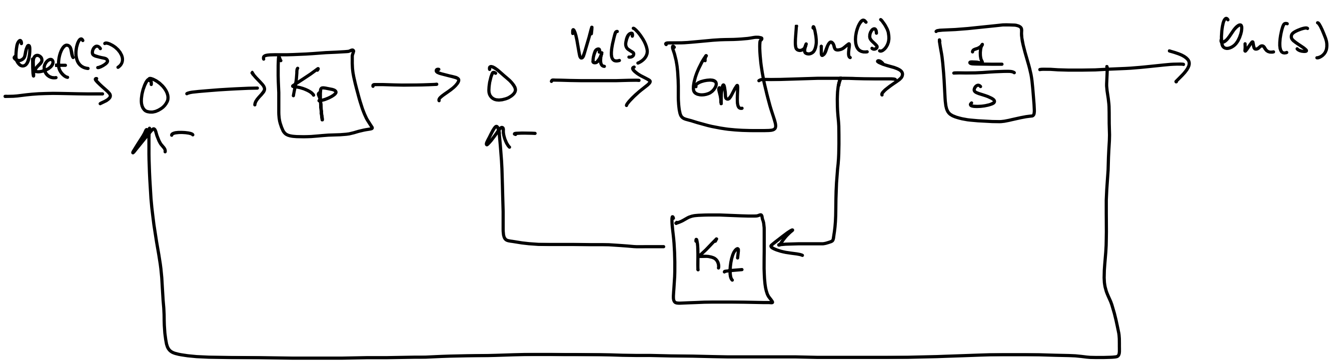
$$\tau_n = \frac{1}{|s|}$$

$$\tau_1 = \frac{1}{|-11016.6|} = 0.0000908 \text{ sec}$$

$$\tau_2 = \frac{1}{|-10981.77|} = 0.0000911 \text{ sec}$$

C. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{ref}}$ *and* there is proportional gain K_p and tachometer gain K_f

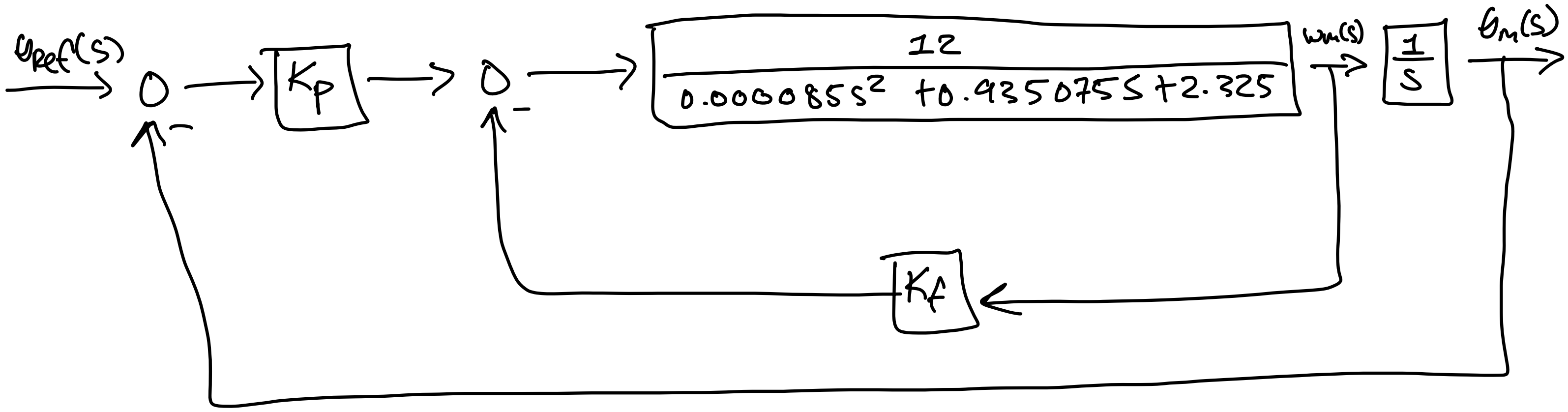
1. Determine the range of K_p and K_f so the system is stable
2. Add a disturbance input to the model and find the open-loop transfer function $\frac{\Theta_m}{T_d}$. Determine values for K_p and K_f to minimize the disturbance



$$G_m = \frac{\omega_m(s)}{V_a(s)} = \frac{K_m}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}$$

$$G_m = \frac{12}{(0.0001)(0.85)s^2 + ((0.0001)(0.75) + (1.1)(0.85))s + (1.1)(0.75) + 12(0.125)}$$

$$G_m = \frac{12}{0.000085\,s^2 + 0.935075\,s + 2.325}$$



2 user set parameters

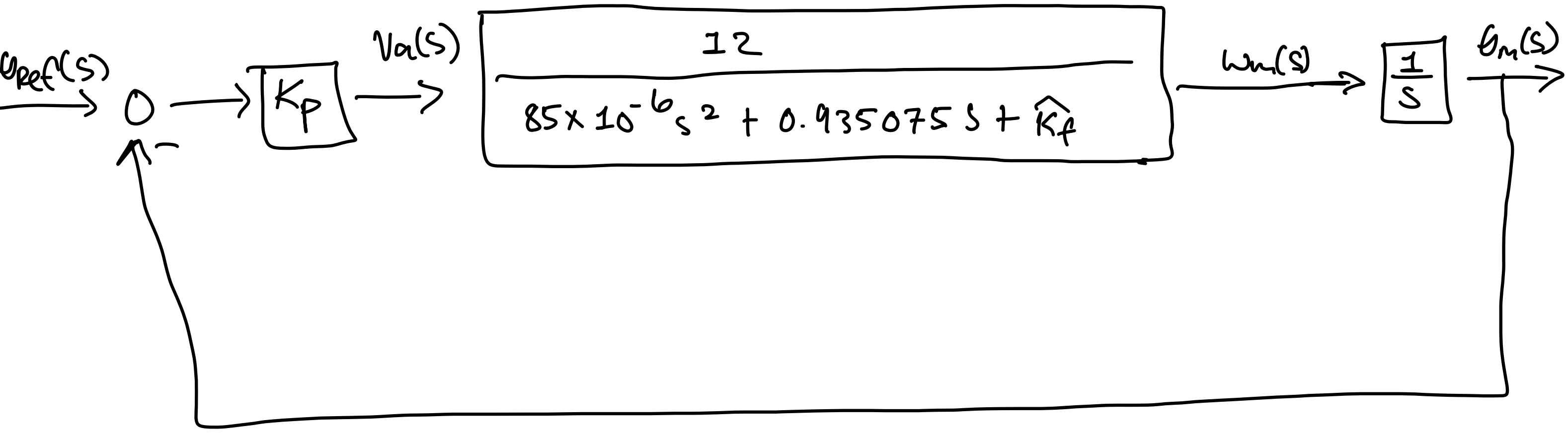
$0 \le K_f < \infty$

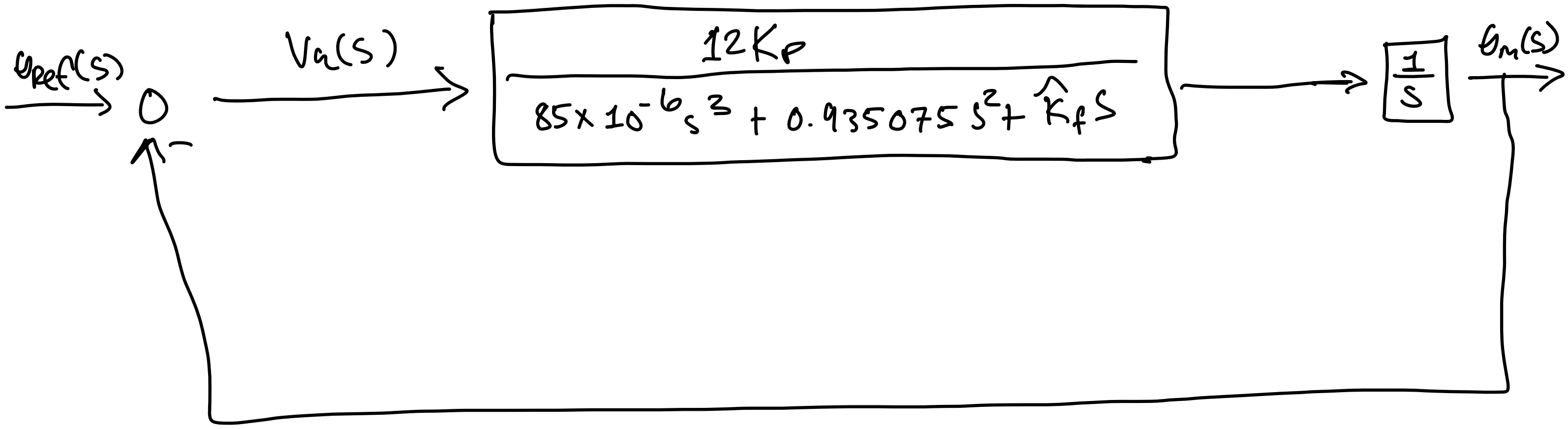
$0 \le K_p < \infty$

$$\frac{\text{Out}}{\text{In}} = \frac{\dfrac{12}{0.000085s^2 + 0.935075s + 2.325}}{1 + \dfrac{12\,K_f}{0.000085s^2 + 0.935075s + 2.325}}$$

$$\frac{\text{Out}}{\text{in}} = \frac{12}{0.000085s^2 + 0.935075s + \underbrace{(2.325 + 12K_f)}_{\hat{K_f}}}$$

$$\frac{\text{Out}}{\text{in}} = \frac{12}{85\times10^{-6}s^2 + 0.935075\,s + \hat{K_f}}$$





$$\frac{\theta_m(S)}{\theta_{ref}(S)} = \frac{\dfrac{12K_p}{85\times10^{-6}s^3 + 0.935075S^2 + \hat{K_f}S}}{1 + \dfrac{12K_p}{85\times10^{-6}s^3 + 0.935075S^2 + \hat{K_f}S}}$$

$$\frac{\theta_m(S)}{\theta_{ref}(S)} = \frac{12Kp}{85\times10^{-6}s^3 + 0.935075S^2 + \hat{K_f}S + 12Kp} = \frac{B(S)}{A(S)}$$

1.) RA

$$\begin{array}{c|cc}
s^3 & 85\times10^{-6} & \hat{K_f} \\
s^2 & 0.935075 & 12K_p \\
s^1 & X_{31} & 0 \\
s^0 & 12Kp &
\end{array}$$

$$X_{31} = \frac{0.935075\hat{K_f} - (12K_p)(85\times10^{-6})}{0.935075\hat{K_f}} > 0$$

$$= 0.935075\hat{K_f} - 0.01224KP > 0$$

$$= 0.935075(2.325 + 12Kf) - 0.01224Kp > 0$$

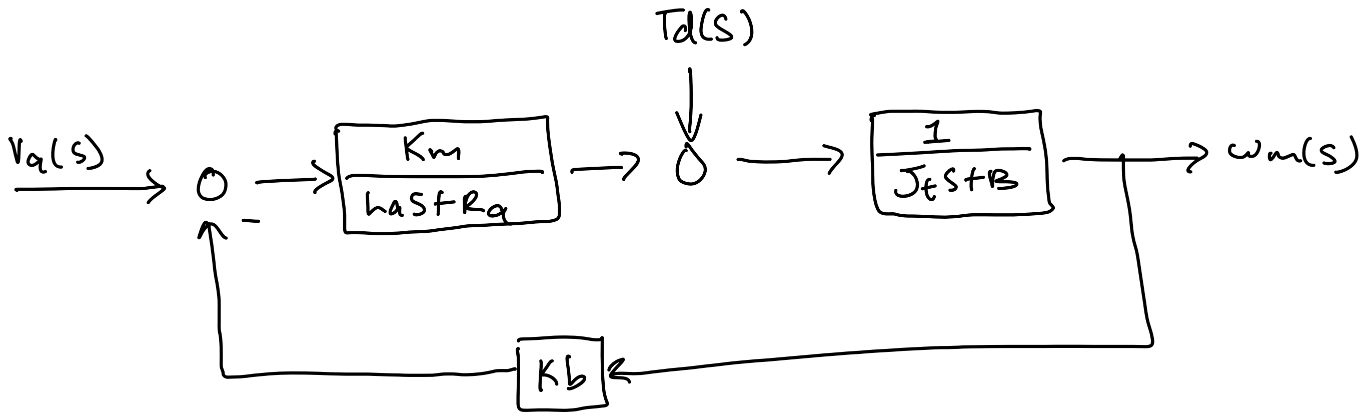
$$= 2.17 + 11.22Kf - 0.01224Kp > 0$$

$$= 11.22Kf - 0.01224Kp > -2.17$$

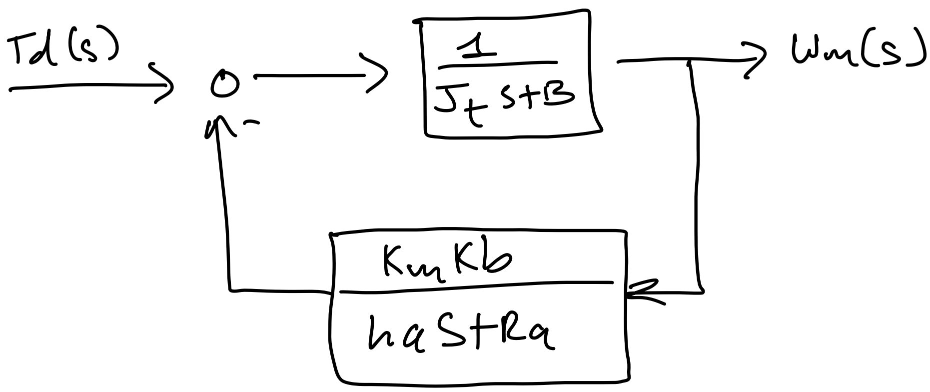
$12K_p > 0$ $\qquad \hat{K}_f = 2.325 + 12K_f > 0$ $\qquad$ 3 Ranges $\Rightarrow$

$\underline{K_p > 0}$ $\qquad \underline{K_f > -0.19}$

2.)



$\Rightarrow$ set $V_a(s) = 0$ to solve for $\dfrac{W_m(s)}{T_d(s)}$



$$\frac{W_m(s)}{T_d(s)} = \frac{\dfrac{1}{J_t s + B}}{1 + \dfrac{K_m K_b}{(L_a s + R_a)(J_t s + B)}} = \frac{\dfrac{1}{J_t s + B}}{\dfrac{(L_a s + R_a)(J_t s + B) + K_m K_b}{(L_a s + R_a)(J_t s + B)}}$$

$$= \frac{L_a s + R_a}{(L_a s + R_a)(J_t s + B) + K_m K_b} = \frac{L_a s + R_a}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}$$

$$\frac{W_m(s)}{T_d(s)} = \hat{G}_m(s) = \frac{L_a s + R_a}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}$$

$$W_m(s) = G_m V_a + \hat{G}_m T_d$$

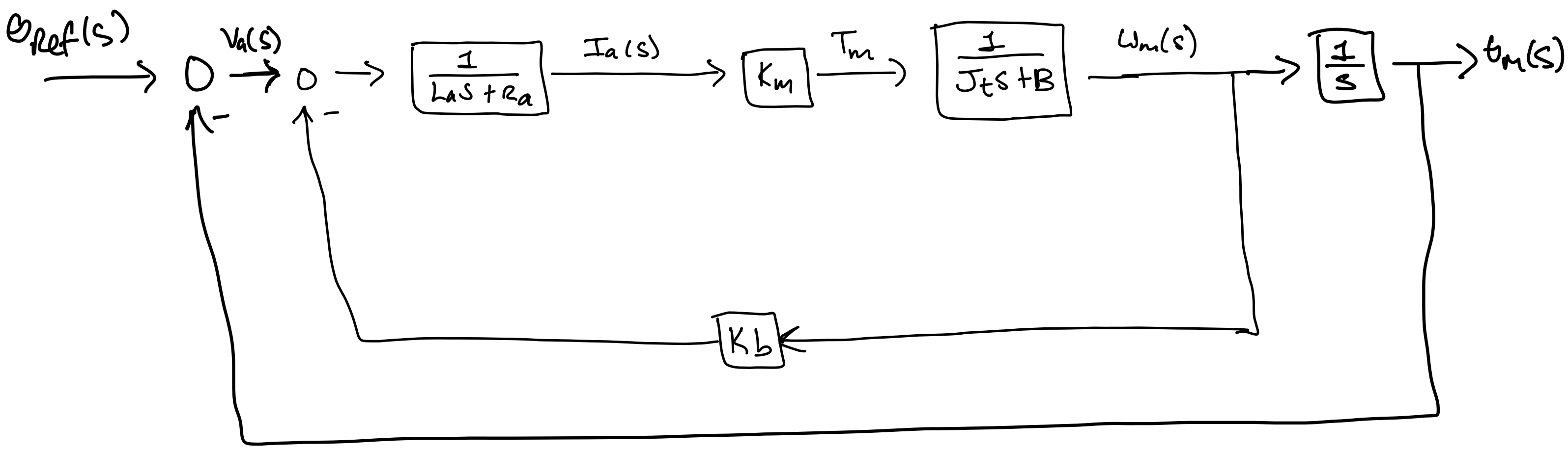
$$W_m(S) = \frac{K_m V_a(S)}{L_a J_t S^2 + (L_a B + R_a J_t)S + R_a B + K_m K_b} + \frac{(L_a S + R_a)\, T_d(s)}{L_a J_t S^2 + (L_a B + R_a J_t)S + R_a B + K_m K_b}$$

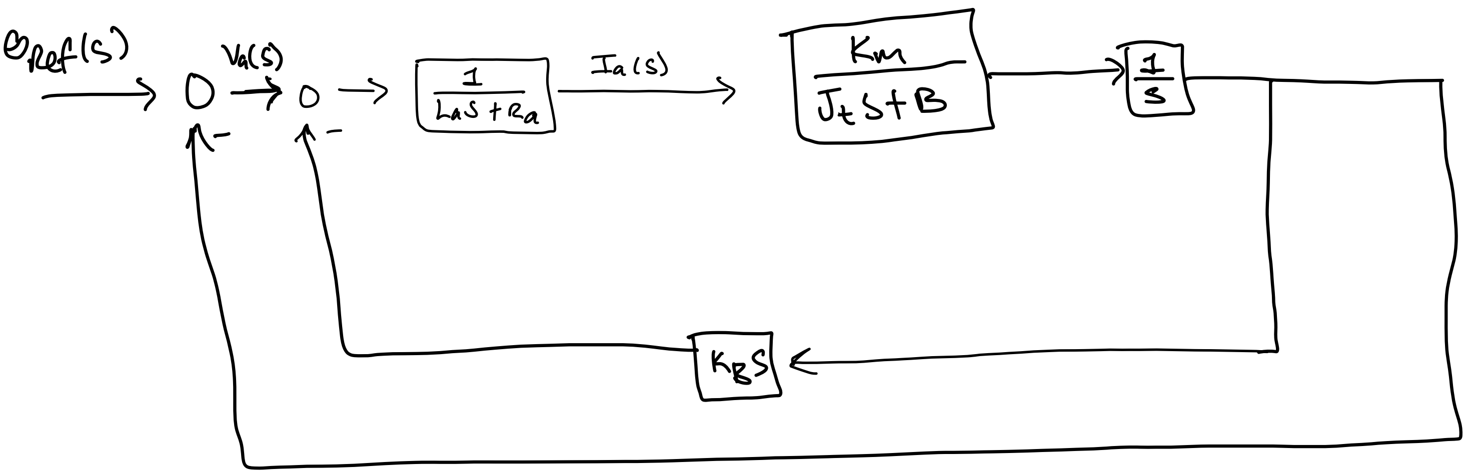
$$\theta_m(S) = W_m(S) \cdot \frac{1}{S} = \frac{W_m(S)}{S}$$

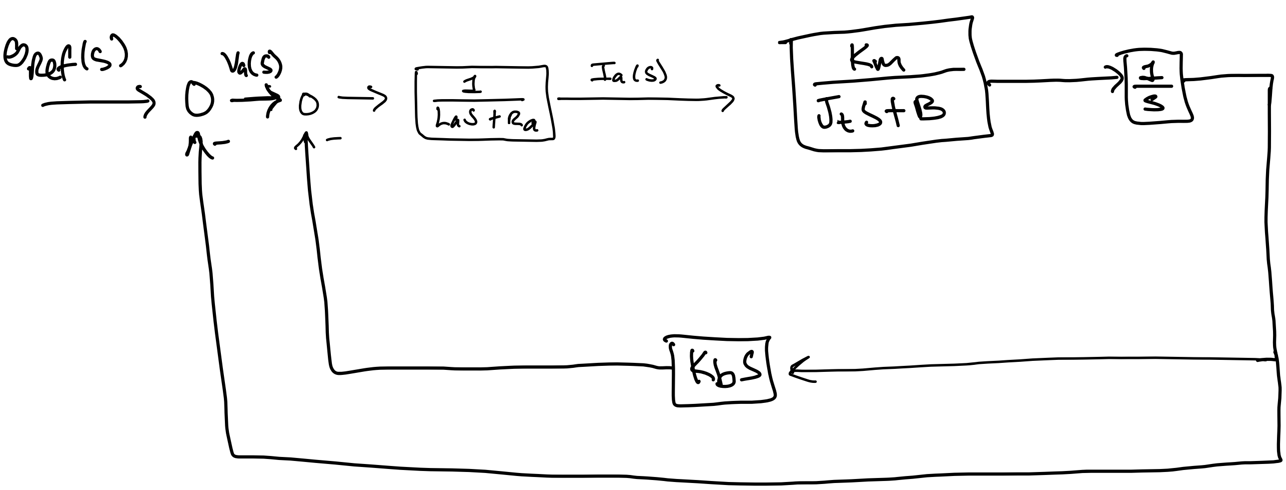
B. When the open-loop transfer function is $\frac{\Theta_m}{\Theta_{\text{ref}}}$

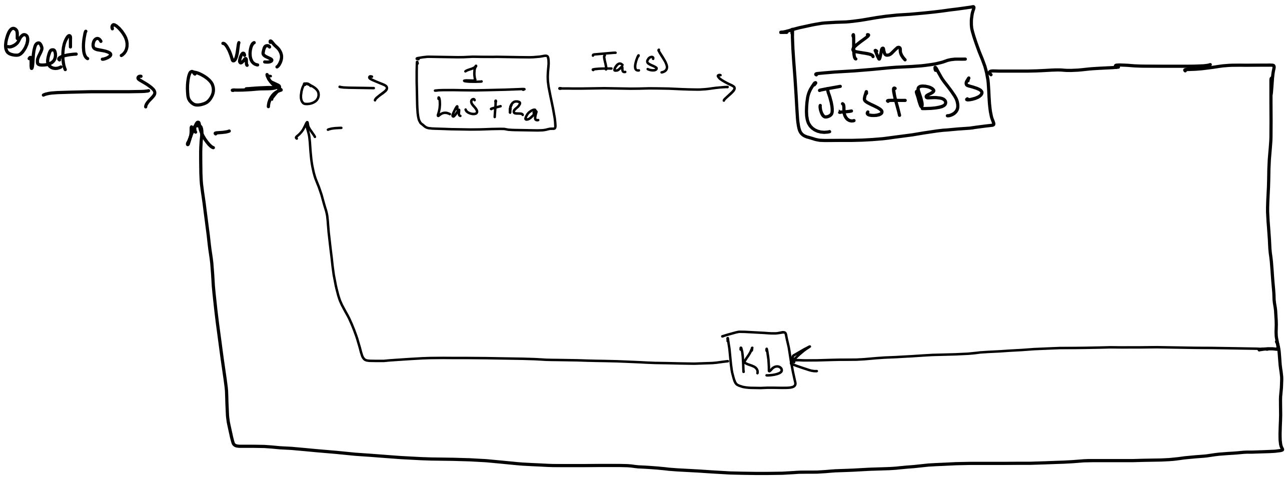
CoRRECT WAY

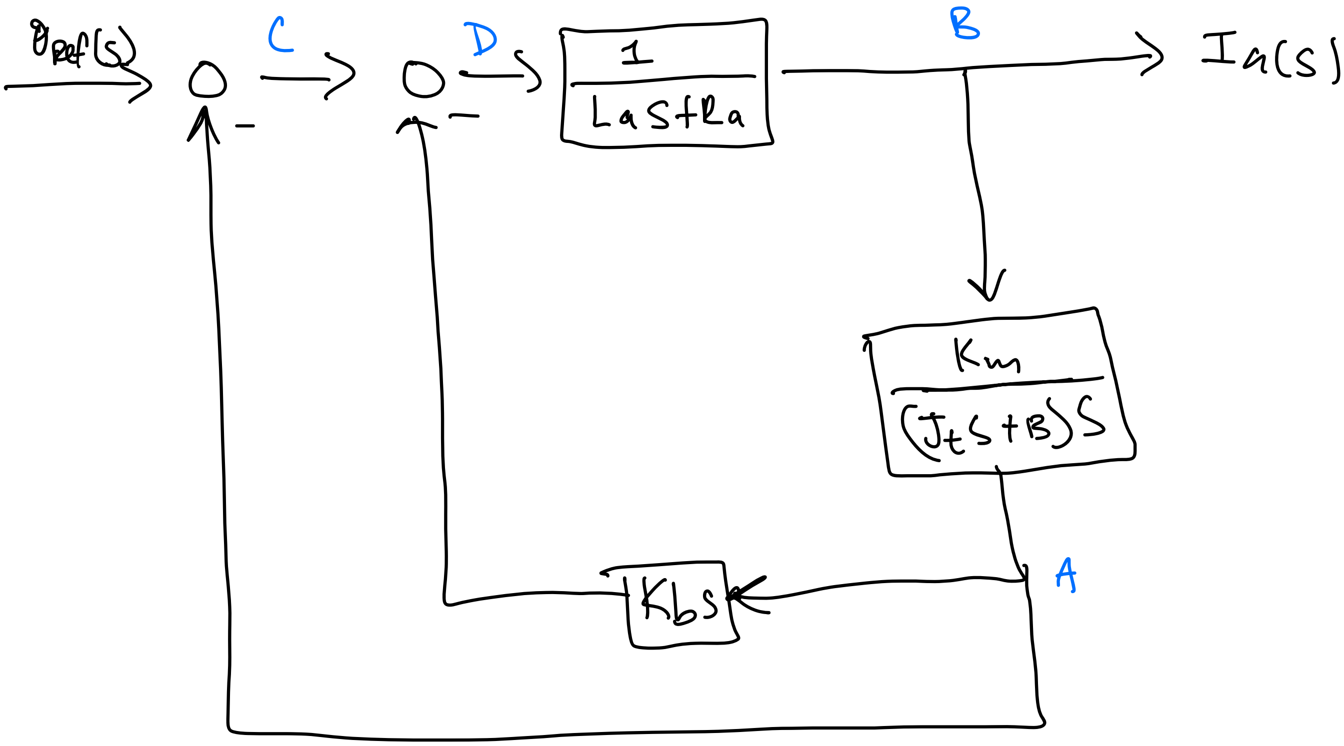
1. The open-loop transfer function, $\frac{I_a}{\Theta_{\text{ref}}}$. What is the steady state current given a setpoint input?
2. The poles and the time constants











$C = \theta_{ref} - A$

$D = C - A K_b S$

$A = \dfrac{K_m}{S(J_t S + B)} \cdot B$

$B = I_a(S)$

$I_a(S) = \dfrac{D}{L_a S + R_a} \Rightarrow D = I_a(s)(L_a S + R_a)$

$D = \theta_{ref} - A - A K_b s$

$$D = \theta_{ref} - \frac{K_m}{s(J_t s + B)} I_a(s) - \frac{K_m}{s(J_t s + B)} I_a(s) K_b s$$

$$I_a(s)(L_a s + R_a) = \theta_{ref} - \frac{K_m}{s(J_t s + B)} I_a(s) - \frac{K_m}{s(J_t s + B)} I_a(s) K_b s$$

$$\theta_{ref} = I_a(s)\left((L_a s + R_a) + \frac{K_m}{s(J_t s + B)} + \frac{K_m K_b s}{s(J_t s + B)}\right)$$

$$\frac{I_a(s)}{\theta_{ref}} = \frac{1}{(L_a s + R_a) + \frac{K_m}{s(J_t s + B)} + \frac{K_m K_b s}{s(J_t s + B)}}$$

- $R_a = 1.1$
- $L_a = 0.0001$
- $K_m = 12$
- $B = 0.75$
- $J_t = 0.85$
- $K_b = 0.125$

$$\frac{I_a(s)}{\theta_{ref}} = \frac{s(J_t s + B)}{(L_a s + R_a)(s)(J_t s + B) + K_m + K_m K_b s}$$

$$\frac{I_a(s)}{\theta_{ref}} = \frac{J_t s^2 + B s}{L_a J_t s^3 + (R_a J_t + L_a B)s^2 + (R_a B + K_m K_b)s + K_m}$$

$$\frac{I_a(s)}{\theta_{ref}} = \frac{0.85 s^2 + 0.75 s}{(0.0001)(0.85)s^3 + ((1.1)(0.85) + (0.0001)(0.75))s^2 + ((1.1)(0.75) + (12)(0.125))s + 12}$$

$\theta_{ref} \rightarrow$ input $\rightarrow$ set input $= \frac{1}{s}$

$$I_a(\infty) = \lim_{s \to 0} s\, I_a(s) = \lim_{s \to 0} s \left[I_a(s) \right] \frac{1}{s} \Bigg|_{s=0} = 0$$

$$A(s) = 0.000085s^3 + 0.935075s^2 + 2.325s + 12$$

$$s = -1.24 + j3.36, \quad -1.24 - j3.36, \quad -10998.4$$

$$\tau_n = \frac{1}{|R|s||}$$

$$\tau_1 = \frac{1}{|-1.24|} = 0.81\ s$$

$$\tau_2 = \frac{1}{|-10998.4|} = 0.000091\ s$$

2. **Given** the input output transfer function:

$$\frac{Y(s)}{R(s)} = \frac{s+126}{(s+6)(s^2+10s+21)} \qquad L = GH$$

$$\frac{1}{\frac{1}{G}+H} \leftarrow \text{divide by numerator}$$

Find:

A. Find the loop gain L, and the **steady state error** for (i) position, (ii) velocity, (iii) and acceleration inputs. $\delta = 1$

B. When the input is an impulse, find:

1. $y(0)$
2. $y(\infty)$ -1
3. $y(1)$ X

A)

$$\frac{Y(s)}{R(s)} = \frac{s+126}{(s+6)(s^2+10s+21)} = \frac{s+126}{s^3+16s^2+81s+126} = \frac{1}{\frac{s^3+16s^2+81s}{s+126}+\frac{126}{s+126}}$$

$$\frac{1}{G} = \frac{s^3+16s^2+81s}{s+126}, \quad G = \frac{s+126}{s^3+16s^2+81s}, \quad H = \frac{126}{s+126}$$

$$L(s) = GH = \left(\frac{\cancel{s+126}}{s^3+16s^2+81s}\right)\left(\frac{126}{\cancel{s+126}}\right) = \frac{126}{s^3+16s^2+81s}$$

$$L(s) = \frac{126}{s^1(s^2+16s+81)} \qquad \text{Type 1}$$

$$K_p = L(0) = \frac{126}{0} \to \infty, \quad e_{Kp} = \frac{1}{1+\infty} = 0$$

$$K_v = \lim_{s\to 0} s\,L(s) = \cancel{s}\left[\frac{126}{\cancel{s}(s^2+16s+81)}\right]\Bigg|_{s=0} = \frac{126}{81}, \quad e_{Kv} = \frac{1}{K_v} = \frac{81}{126}$$

$$K_A = \lim_{s\to 0} s^2 L(s) \quad s^{\cancel{2}}\left[\frac{126}{\cancel{s}(s^2+16s+81)}\right]\Bigg|_{s=0} = \frac{0}{81} = 0, \quad e_{KA} = \frac{1}{K_A} = \frac{1}{0} = \infty$$

B)

$$\frac{Y(s)}{R(s)} = \frac{s+126}{(s+6)(s+3)(s+7)} = -\frac{40}{s+6} + \frac{41}{4(s+3)} + \frac{119}{4(s+7)}$$

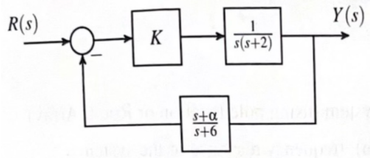
$$y(t) = -40e^{-6t} + \frac{41}{4}e^{-3t} + \frac{119}{4}e^{-7t}$$

$y(0) = 0$

$y(\infty) = 0$

$y(1) = 0.438$

3. Given the control system below where $0 < K < \infty$



Find:

A. Sketch the Root Locus for the system when $\alpha = 4$

B. Determine the range of K such that the system is stable when $\alpha = 4$

C. Determine the range of α such that the system is stable when $K = 10$

A) $L(s) = KGH = \dfrac{K(s+4)}{s(s+2)(s+6)} = \dfrac{K(s+4)}{s^3+8s^2+12s}$ lg zeros: -4

lg poles: $0, -2, -6$

$m=1 \quad n=3$

$\sigma_i = \dfrac{(0-2-6)-(-4)}{3-1} = \dfrac{-4}{2} = -2 \qquad K = |3-1|-1 = 1 \qquad K = \{0, 1\}$

$\theta_0 = \dfrac{(2\cdot 0+1)\pi}{3-1} = \dfrac{\pi}{2} \qquad \theta_1 = \dfrac{(2\cdot 1+1)\pi}{3-1} = \dfrac{3\pi}{2}$

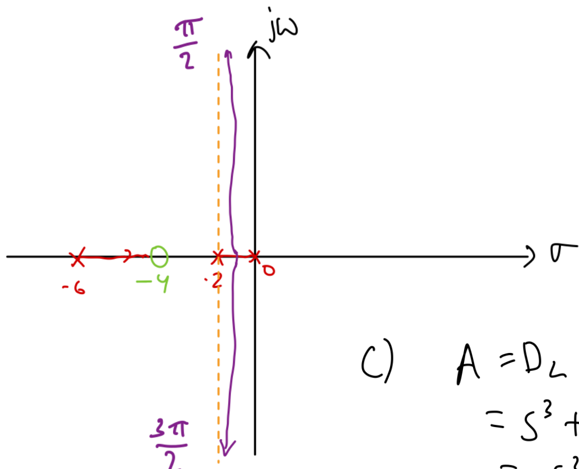
$\dfrac{BA}{BI}$: $D_L N_L' - D_L' N_L = 0$

$= (s^3+8s^2+12s)(1) - (3s^2+16s+12) = -2s^3-20s^2-64s-48$

$s = -4.47 \pm j1.59, \ -1.07 \leftarrow BA$

$K_{BA} = \dfrac{-D_L}{N_L} = \dfrac{-s^3+8s^2+12s}{s+4}\Bigg|_{BA=-1.07} = 1.67$

$K_{cr} = 0$



B) $A = D_L + KN_L$

$= s^3+8s^2+12s + K(s+4)$

$= s^3+8s^2+(12+K)s+4K$

s^3	1	$12+K$
s^2	8	$4K$
s^1	X_{31}	0
s^0	X_{41}	

$X_{31} = \dfrac{(8)(12+K) - 4K}{8}$

$= \dfrac{96+4K}{8} > 0 \Rightarrow K > -24$

$X_{41} = \dfrac{X_{31}(4K) - 0}{X_{31}} = 4K > 0$

$K > 0$

C) $A = D_L + K N_L$

$= s^3+8s^2+12s+10(s+\alpha)$

$= s^3+8s^2+22s+10\alpha$

s^3	1	22
s^2	8	10α
s^1	X_{31}	0
s^0	X_{41}	

$X_{31} = \dfrac{(8)(22) - 10\alpha}{8} = \dfrac{176-10\alpha}{8} > 0$

$176 - 10\alpha > 0$

$\alpha < 17.6$

$X_{41} = \dfrac{(X_{31})(10\alpha) - 0}{X_{31}} = 10\alpha > 0$

$\alpha > 0$

$0 < \alpha < 17.6$